



M Rupesh Kumar Yadav
Geoinformatics
Indian Institute of Technology Bombay

24M0319
M.Tech.
Gender: Male
DOB: 21/03/2003

Examination	University	Institute	Year	CPI / % Credits
Post Graduation	IIT Bombay	IIT Bombay	2026	9.45 Core: 72 Total: 120
Graduation	NIT, Andhra Pradesh	NIT, Andhra Pradesh	2024	8.56
Graduation Specialization: Computer Science and Engineering				
Intermediate	BIEAP	Sri Chaitanya Junior College	2020	9.86
Matriculation	BSEAP	Narayana E.M High School	2018	10

SCHOLASTIC ACHIEVEMENTS

- Achieved **97** Percentile in GATE Computer Science and Information Technology.

PUBLICATIONS

- Enhancing Hyperspectral Imagery Classification using CNN and **Vision Transformer**, submitted to **PReMI 2025**.
- Co-authored, **Graph-Based Stance Grouping in Multi-Participant Discussions** published at ICSICE 2024.
- Quantum Convolutional Neural Networks for Slum Classification using Sentinel-2 Imagery**, under review at Quantum Information Processing (**QIP**) Journal 2025.

PROFESSIONAL EXPERIENCE

- Intern | Center for Artificial Intelligence and Robotics (CAIR), DRDO** [May'23-Jul'23]
 - Developed and integrated YOLO models(v7,v8,Complex) for 2D/3D object detection using camera & LiDAR data.
 - Utilized **ROS2** to integrate and deploy custom detection models for **real-time**, on-vehicle testing and validation.
 - Achieved a mAP of **0.865** across diverse weather conditions by fine-tuning models on a proprietary in-house data.
- Machine Learning Intern | Hindustan Aeronautics Ltd (HAL)** [May'22-July'22]
 - Engineered a **predictive** maintenance model leveraging **13 years of SQMS data** to forecast snag dispositions.
 - Attained **98.76%** testing accuracy by benchmarking a suite of ML algorithms including SVM and XGBoost etc.
 - Delivered analytics dashboards** reducing manual reporting time by **80%** and enhance **workflow efficiency**.

KEY PROJECTS

- AI models for Urban Informal Settlement Detection | M.Tech Project** [Jun'25-Present]
Guide: Prof. Avik Bhattacharya
 - Benchmarked **OBIA** & machine learning models (Random Forest, XGBoost) for slum detection using Sentinel-2.
 - Developed Deep Learning pipelines using CNNs and **Quantum CNNs** for urban informal settlement classification.**Future Work:**
 - Utilize **Small Vision-Language Models (sVLMs)** like Presto for slum detection & domain generalization tasks.
 - Integrate LLMs for **advanced reasoning** and **contextual interpretation**, by incorporating geospatial context.
- Quantum-CNN for Image Classification | M.Tech Seminar | Guide: Prof. B. Krishna Mohan** [Jan'25-May'25]
 - Implemented NEQR, FRQI, & MCQI quantum image encoding techniques for multi-band remote sensing imagery.
 - Built MQCNN for Sentinel-2 LULC classification, delivering **5%** higher accuracy than a comparable CNN baseline.
- Deep Learning for Hyperspectral Imagery Classification | BTech Project** [Feb'24-Apr'24]
| Guide: Asst. Prof. Nagesh Bhattu Sristy | NIT Andhra Pradesh
 - Benchmarked diverse architectures (SVM, 3DCNN, GCNs, Autoencoder, Transformers) to guide final model design.
 - Engineered a novel, **transformer**-based model for hyperspectral classification in a low-data regime (10% training).
 - Achieved state-of-the-art accuracy, including **99.80%** on the Indian Pines and **97.54%** on the PaviaU datasets.
 - First-authored a research paper on the model, presented at ICICA-2024 conference, submitted to **PReMI 2025**.
- ProdHub: AI-Powered Personal Productivity Hub | Self Project** [Jul'25-Aug'25]
 - Architected a serverless productivity hub using **React**, **Firestore** and **gcloud** for integrated project management.
 - Integrated the Google **Gemini API** to power an AI engine for smart scheduling and context-aware task suggestions.
 - Developed an **AI based** goal decomposition feature to automatically structure high-level goals into project plans.
 - Enabled a **unified** schedule view by securely syncing Google Calendar data using secure, read-only **OAuth 2.0**.
- Intelligent Multi-Agent Knowledge Assistant | Self Project** [Jun'25-Aug'25]
 - Engineered a multi-agent system using LangChain & **Gemini** to route queries to specialized HR and Tech agents.
 - Implemented a dual-source **Hierarchical RAG** pipeline with **FAISS**, optimizing information retrieval by using **Parent-Child chunking**, Query Expansion, and **Flashrank re-ranking**, along with external python,web tools.
 - Built a robust agentic workflow with **Guardrail** & **Triage** layers, ensuring reliable, domain-specific query handling.
- Semantic Paraphrase Identification | Self Project** [Sep'24-Nov'24]
 - Developed a **Siamese LSTM network** for NLP task of semantic paraphrase detection using the MRPC dataset.
 - Enhanced **semantic representation** using pre-trained **GloVe embeddings**, achieving **91.4% test accuracy**.
 - Built an **end-to-end** pipeline with **transfer learning** and Early Stopping, integrating **MLflow for tracking**.

- Pansharpening-using-PCA-for-Satellite-Imagery** | *Course Project* [Nov'24-Dec'24]
GNR607 | Guide: Prof. B. Krishna Mohan
 - Implemented a PCA-based **pansharpening** algorithm to enhance the spatial resolution of multispectral imagery.
 - Developed a data pipeline to automatically extract and process satellite imagery from Google Earth Engine (**GEE**).
 - Deployed the image enhancement model as an interactive web application using **Flask** with **PostgreSQL** database.
- Emergency Services & Flood Zones Locator** | *Self Project* [Oct'24-Nov'24]
 - Developed cross-platform disaster response app with flood modeling, **PostGIS** analysis, and live emergency data.
 - Implemented **FastAPI** backend with **GiST** indexing and **Redis caching**, reducing location query times by **99%**.
 - Developed a responsive **Leaflet.js** map with interactive filters, live geolocation, and deployed on **google cloud**.
- GeoWebClient: Interoperable GIS Web Application** | *Course Project* | *GNR629* [Jan'25-Feb'25]
Advances in Geospatial Standards, Interoperability and Knowledge Discovery | Guide: Prof. Surya S.Durbha
 - Developed a **web client** for WMS, WFS & WCS web services using **React**, enhancing data retrieval & display.
 - Configured a full-stack GIS server using **PostGIS**, GeoServer, and Tomcat to serve spatial data via OGC services.
 - Implemented **XML parsing**, **spatial querying**, **multi-layer support** and integration with **external servers**
- SOS WebClient: A Sensor Data Visualization Interface** | *Course Project* [Mar'25-Apr'25]
GNR629 | Guide: Prof. Surya S.Durbha
 - Built a lightweight webclient to visualize real-time data from OGC Sensor Observation Service (**istSOS**) endpoints.
 - Implemented real-time sensor discovery, interactive map-based filtering to streamline data exploration & analysis.
- Image Super-Resolution** | *Self Project* [Feb'25-Mar'25]
 - Designed a lightweight CNN in PyTorch for **4× image super-resolution** on 4,500 gaming-scene LR–HR pairs.
 - Applied **residual learning**, **PixelShuffle** upsampling, & **mixed-precision** resulting 0.56 score on joint metric.
 - Converted & deployed the model using **TorchServe** & **ONNX** for efficient inference; evaluated via PSNR, SSIM.
- Pedestrian Tracking using DeepSort & YOLO** | *Self Project* [Mar'24-Apr'24]
 - Engineered and deployed a modular end-to-end **YOLOv5 + DeepSort** pedestrian tracking pipeline via **FastAPI**.
 - Achieved 87% re-identification accuracy across different camera views using a **ResNet-50** feature extraction model.
- RIBS Identification, Labelling and Segmentation using FPN** | *Self Project* [Feb'22-Apr'22]
 - Built a deep learning pipeline for rib segmentation and labeling in chest X-rays using U-Net++ (EfficientNet-B0).
 - Achieved a **Dice score** of **0.834** on the VinDr-RibCXR dataset, validated model performance on JSRT dataset.
- Time Series Forecasting for Financial Market** | *Self Project* [Nov'23-Dec'23]
 - Performed **EDA** on stock market data, visualizing closing prices to uncover patterns, trends, and seasonality.
 - Applied **ADF**, **hypothesis tests**, **ACF**, **PACF**, and **decomposition** to assess stationarity and data quality.
 - Built ARIMA, SARIMA, & LSTM models, with stateful-LSTM achieving best performance (RMSE:10.3, R²:0.83).
- Financial News Sentiment Analysis** | *Self Project* [Aug'23-Sep'23]
 - Leveraged the Financial **PhraseBank dataset** (4,840 expert-annotated sentences) for **sentiment classification**.
 - Benchmarked BERT & DistilBERT, achieving best performance with fine-tuned **FinBERT**(Acc: 90.9%, F1: 0.91).
- Triple GAN** | *Self Project* [Feb'23-Apr'23]
 - Implemented the TripleGAN architecture to generate synthetic MNIST images, reaching a peak accuracy of **99.1%**.

Academic Courses	
Advanced Database Systems Deep Learning for Vision Big Data Engineering Applied Machine Learning Natural Language Processing Social Network Analytics Affective Computing Human Resource Management Soft Skills	

TECHNICAL SKILLS	
Programming Languages	Python, C/C++, Java, Javascript, SQL
AI & Data Science	Tensorflow, Pytorch, OpenCV, Scikit-learn, NLTK, spaCy, DVC, MLFlow, Langchain, OpenAI SDK, Hugging Face, Matplotlib
Web/App Development	Node.js, React, Express.js, Flask, Streamlit, FastAPI
Tools & Platforms	Linux, Git, MySQL, MongoDB, Tableau, PowerBI, ArcGIS, PostgreSQL, Apache Airflow, Docker, GCP, Azure, BigQuery, PySpark

POSITION OF RESPONSIBILITY	
● General Secretary Taskforce NIT Andhra Pradesh [Aug'23–Apr'24] ○ Led 10+ institute-wide initiatives with 45+ members, aligning with national programs (ODOP, Swachh Bharat 3.0); organized 1,500+ participant events. Directed flagship technical & cultural festivals, managing 15 cross-club teams, Rs.2.5M+ budgets, and 40+ major events. ● Joint Secretary Computer Science and Engineering Association (CSEA) NIT AP [Aug'22–May'23] ○ Spearheaded CSEA initiatives (workshops, webinars, coding/design contests, TECH EXPO), engaging 500+ participants & industry experts; headed design, coding & management teams, & served as acting secretary.	

Hobbies
Swimming Sudoku Badminton Photography Gaming Blogging Squash